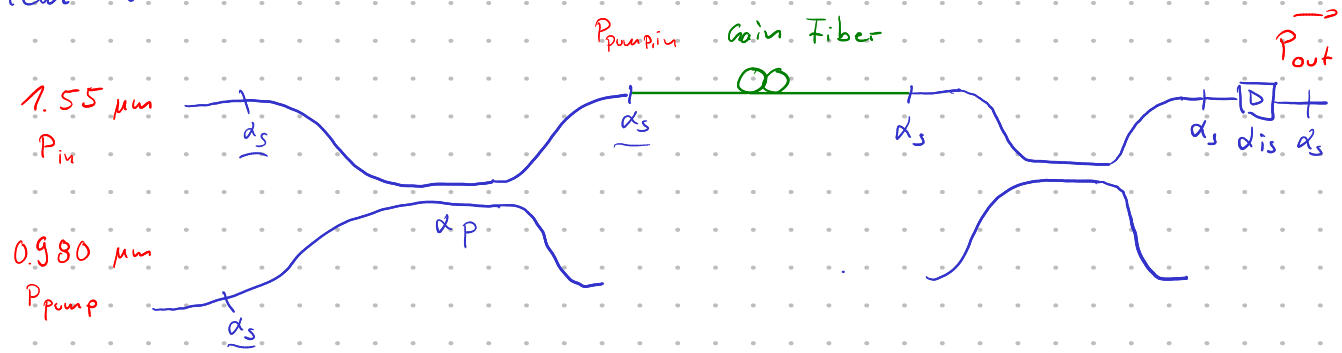
AOFC Exercise 10

Problem 17

a)



$$G_{fiber}\Big|_{dB} = g_{pump} \left[\frac{dB}{mW}\right] \cdot P_{pump,in}$$

$$P_{pump,in}\Big|_{dB} = P_{pump}\Big|_{dBm} - 2\alpha_s - \alpha_p$$

$$= 10 \cdot \log_{10} \frac{30\,mW}{1\,mW} - 2 \cdot 0.1\,dB - 0.45\,dB = 14.12\,dBm$$

$$P_{pump,in} = 10^{14.12\,dBm/10} = 25.82\,mW$$

$$G_{fiber}\Big|_{dB} = 1.2\,dB/mW \cdot 25.82\,mW = 31\,dB$$

$$G_{total} = P_{out} - P_{in} = G_{fiber}\Big|_{dB} - 5 \cdot \alpha_s - \alpha_{is} = 31\,dB - 5 \cdot 0.1\,dB - 1\,dB$$

$$= 29.5\,dB$$

Gain of the whole device!

c) $P_{in} = -30\,dBm$ $\qquad P_{out} = P_{in} + G_{total}$

$$P_{out}\Big|_{dB} = -30\,dBm + 29.5\,dB = -0.5\,dBm$$

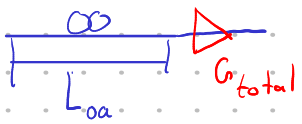
$$\hat{=} 0.98\,mW$$

d) n_{sp} -> spontaneous emission factor

$n_{sp} = 0$ -> perfect inversion (ideal amplifier)

$n_{sp} > 1$ -> incomplete inversion

$n_{sp} \approx 2$ -> typical amplifier (EDFA!)



noise figure $F = \frac{SNR_{in}}{SNR_{out}} = \frac{P_{in}/P_{noise,in}}{P_{out}/P_{noise,out}}$

$= \frac{P_{in}}{P_{out}} \cdot \frac{P_{noise,out}}{P_{noise,in}} = \frac{1}{G} \cdot \frac{P_{noise,out}}{P_{noise,in}}$

Fiber: $P_{noise,in} = P_{noise,out}$ -> no additional noise contribution from the fiber!

$F_{fiber} = \frac{1}{G_{fiber}}$ with $0 < G_{fiber} < 1$

Amplifier: $P_{noise,out} = P_{noise,in} + 4 \cdot \underbrace{S_{ASE} \cdot \Delta f}_{\text{noise power per bandwidth}}$ -> ASE noise spectral density

ASE = amplified spontaneous emission

with $P_{noise,in} = 2 \cdot \Delta f \cdot hf$ $\quad S_{ASE} = n_{sp} \cdot hf \,(G_{amp} - 1)$

$F_{amp} = \frac{1}{G_{Amp}} \cdot \left(\frac{4 \cdot S_{ASE} \cdot \Delta f}{2 \cdot \Delta f \cdot hf} + 1 \right)$

$= \frac{1}{G_{amp}} \left(\frac{2 \cdot n_{sp} \cdot hf \,(G_{amp} - 1)}{hf} + 1 \right)$

$F_{amp} = \frac{1}{G_{amp}} \left(2 \cdot n_{sp} \cdot (G_{amp} - 1) + \right) = 2 \cdot n_{sp} \left(1 - \frac{1}{G_{amp}} \right) + \frac{1}{G_{amp}}$

$$F_{seg} = F_{fiber} + \frac{F_{amp}^{-1}}{G_{fiber}} = \frac{1}{G_{fiber}} + \frac{2 \cdot n_{sp}\left(1 - \frac{1}{G_{amp}}\right) + \frac{1}{G_{amp}}}{G_{fiber}}$$

$$G = G_{amp} = \frac{1}{G_{fiber}}$$

$$F_{seg} = G + 2 n_{sp}(G-1) + 1 - G = 1 + 2 n_{sp}(G-1)$$

$$G = \frac{1}{G_{fiber}} = \frac{1}{e^{-2\alpha \cdot L_{oa}}} = e^{2\alpha \cdot L_{oa}} = 10^{\alpha_{dB} \cdot L_{oa}/10}$$

$$= 10^{29.5\,dB/10}$$

$$= 891 \rightarrow \text{linear}$$

$$F_{seg} = 1 + 2 \cdot 2 (891 - 1) = 3561 \hat{=} 35{,}5\,dB$$

$$G_{total}\Big|_{dB} = \alpha_{dB} \cdot L_{oa} \qquad L_{oa} = \frac{29{,}5\,dB}{0{,}2\,\frac{dB}{km}} = 147.5\,km$$

e) $$F_{tot} = 1 + N(F_{seg} - 1) = 1 + N(G \cdot F_{amp} - 1)$$

$$= 1 + NG \cdot F_{amp} - N \approx NG \cdot F_{amp}$$

$$OSNR\Big|_{mW} = \frac{P_o}{F_{tot} \cdot h \cdot f \, \Delta f} = \frac{P_o}{N \cdot G \cdot F_{amp} \cdot h f \cdot \Delta f} \rightarrow 58\,dB$$

$\lambda = 1.55\,\mu m$

Δf 12.5 GHz

$$OSNR\Big|_{dB} = 58\,dB + P_o\Big|_{dBm} - F_{amp} - \underbrace{\alpha_{dB} \cdot L_{oa}}_{G|_{dB}} - 10 \cdot \log_{10} N$$

$$10 \cdot \log_{10} N = 58\,dB + P_o\Big|_{dBm} - F_{amp} - \alpha_{dB} \cdot L_{oa} - OSNR\Big|_{dB}$$

$$F_{amp} = 2 \cdot n_{sp}\left(1 - 10^{-\alpha_{dB} \cdot L_{oa}/10}\right) + 10^{-\alpha_{dB} \cdot L_{oa}/10}$$

$$= 2 \cdot 2\left(1 - 10^{-0.2\,dB/km \cdot 30km/10}\right) + 10^{-0.2\,dB/km \cdot 30km/10}$$

$F_{amp} = 3.25 \hat{=} 5,12\,dB$

$10 \cdot \log N = 58\,dB + 0\,dBm - 5,12\,dB - 0.2\,dB/km \cdot 30\,km - 30\,dB$

$10 \cdot \log N = 16.88 \rightarrow N = 10^{16.88/10} = 48,75 \rightarrow 48$

$L_{absolute} = 48 \cdot 30\,km = 1440\,km$

Problem 13

a) 50 WDM-Channels, 10 fiber modes

bit rate 200 Tbit/s

$$R_{b\ \text{mode \& WDM}} = \frac{200\ \text{Tbit/s}}{50 \cdot 10} = 400\ \text{Gbit/s}$$

b) $L_{oa} = 100\ \text{km}$, $P_0 = 10\ \text{mW}$, $F_{amp} = 4$, $\alpha_{dB} = 0.2\ \text{dB/km}$

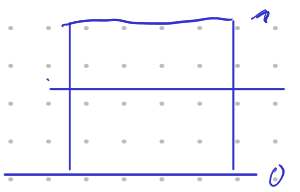
$L = 6600\ \text{km}$

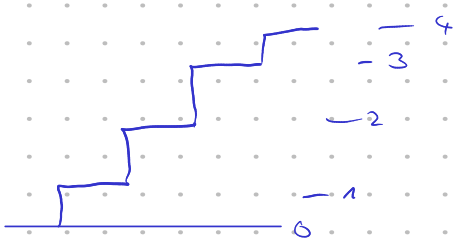
$$OSNR\Big|_{dB} = 58\ \text{dB} + P_0\Big|_{dBm} - F_{amp}\Big|_{dB} - \alpha_{dB} \cdot L_{oa} - 10 \log \frac{L}{L_{oa}}$$

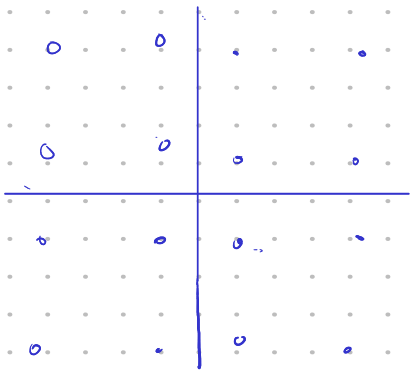
$$= 58\ \text{dB} + 10 \cdot \log \frac{10\ \text{mW}}{1\ \text{mW}} - 10 \log 4 - 0.2\ \text{dB/km} \cdot 100\ \text{km} - 10 \log \frac{6600\ \text{km}}{100\ \text{km}}$$

$$OSNR\Big|_{dB} = 23.78\ \text{dB}$$

c) On-OFF-keying







c) $1 \cdot 10^{-3}$ forward error correction limit, 7 % overhead

$$R_s = R_b \cdot (1 + OH) \cdot \frac{1}{2} \cdot \frac{1}{\log_2 M} \quad \text{.... symbol rate M-QAM}$$

From Diagramm 16 QAM $\quad R_s = 400\,\text{Gbit/s} \cdot (1 + 0.07) \cdot \frac{1}{2} \cdot \frac{1}{\log_2 16}$

$$R_s = 53{,}5\,\text{Gbd}$$

d) $SNR_{req,dB} = 16.5\,\text{dB}$ -> from Diagramm, 16 QAM, BER 1×10^{-3}

$$OSNR = \frac{\Delta f}{12.5\,\text{GHz}} \cdot SNR \;\hat{=}\; OSNR|_{dB} = 10 \cdot \log \frac{\Delta f}{12.5\,\text{GHz}} + SNR_{dB}$$

$R_s \approx \Delta f = 53{,}5\,\text{GHz}$

$$\Rightarrow \quad OSNR_{req,dB} = 10 \cdot \log \frac{53{,}5\,\text{GHz}}{12{,}5\,\text{GHz}} + 16.5\,\text{dB} = 22{,}81\,\text{dB} < 23{,}78\,\text{dB}$$

✓ from before!

$$\Delta f_{max} = 12.5\,\text{GHz} \cdot 10^{(OSNR|_{dB} - SNR_{req}|_{dB})/10}$$

$$\approx 12{,}5\,\text{GHz} \cdot 10^{(23{,}78\,\text{dB} - 16.5\,\text{dB})/10} = 67{,}13\,\text{GHz} \;\hat{=}\; 67{,}13\,\text{Gbd}$$

e) $OSNR\big|_{dB} = 58\,dB + 10 \cdot \log\left(\frac{10\,mW}{1\,mW}\right) - 10 \cdot \log 4 - 0.2\,dB/km \cdot 120\,km - 10 \cdot \log \frac{6600\,km}{120\,km}$

$= 20,6\,dB < OSNR_{req}\big|_{dB} = 22,81\,dB$

↳ Link not working with 16-QAM

from Diagramm: 4-QAM at $SNR_{req}\big|_{dB} = 9,8\,dB$

$R_S = 400\,Gbit/s \cdot (1+0.07) \cdot \frac{1}{2} \cdot \frac{1}{\log_2 4} = 107\,GBd \hat{=} \Delta f = 107\,GHz$

$OSNR_{req}\big|_{dB} = 10 \cdot \log \frac{107\,GHz}{12,5\,GHz} + 9,8\,dB = 19,12\,dB < OSNR\big|_{dB} = 20,6\,dB$

→ transmission link is working by extending bandwidth usage!